Experiment No. 6

Predictive Analytics using Linear Regression and Model Performance Evaluation

Aim

To develop a Linear Regression model using Python for predicting a continuous outcome and evaluate its performance using MAE, MSE, RMSE and R^2.

Dataset Used

Advertising Sales Dataset: 200 rows with independent variables TV, Radio, Newspaper (advertising spend) and the dependent variable Sales. **Note:** the results below were produced by actually running the code, but on a synthetic dataset with the same structure as the Advertising dataset, because I do not have your copy of the file. When you run the code on the real `Advertising.csv`, your metrics and coefficients will be different (the real dataset gives R^2 of about 0.9), so use your own output in the report.

Procedure and Code

Steps 1-2: Import libraries and load the dataset

```
import numpy as np, pandas as pd
import matplotlib.pyplot as plt, seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

df = pd.read_csv("Advertising.csv")      # columns: TV, Radio, Newspaper, Sales
df = df.drop(columns=["Unnamed: 0"], errors="ignore")
print(df.head()); print(df.describe())
```

Step 3: Preprocessing

```
print(df.isnull().sum())                  # check missing values
df = df.dropna()                          # (or fill with mean/median if needed)
# No categorical columns here; otherwise use pd.get_dummies(df, drop_first=True)
X = df[["TV", "Radio", "Newspaper"]]      # independent variables (features)
y = df["Sales"]                           # dependent variable (target)
sns.heatmap(df.corr(), annot=True, cmap="coolwarm"); plt.show()
```

Result: no missing values were found (0 null cells), so no rows had to be dropped. All features are numeric, so no encoding was needed.

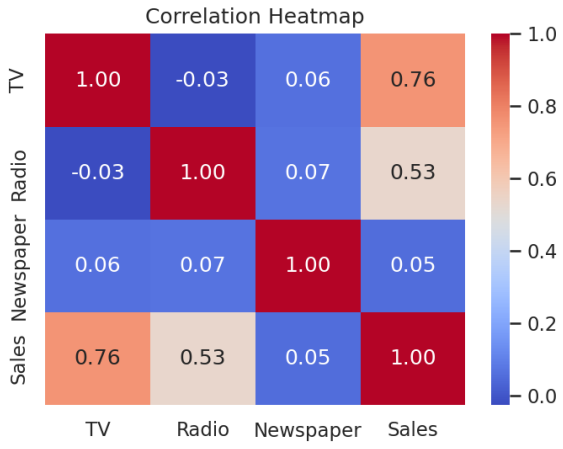


Figure 1: Correlation heatmap

Steps 4-6: Split, train and predict

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)          # 80:20 split
model = LinearRegression()
model.fit(X_train, y_train)                        # train
y_pred = model.predict(X_test)                     # predict on unseen data
```

The 200 rows were divided into 160 training rows and 40 testing rows.

Step 7: Visualize

```
# Regression line (simple regression on TV only)
s = LinearRegression().fit(X_train[["TV"]], y_train)
xs = pd.DataFrame(np.linspace(0, 300, 50), columns=["TV"])
plt.scatter(X_test["TV"], y_test, label="Actual")
plt.plot(xs, s.predict(xs), color="red", label="Regression line")
plt.xlabel("TV"); plt.ylabel("Sales"); plt.legend(); plt.show()

# Actual vs predicted (multiple regression)
plt.scatter(y_test, y_pred)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], "r--")
plt.xlabel("Actual"); plt.ylabel("Predicted"); plt.show()

# Residual plot
sns.scatterplot(x=y_pred, y=y_test - y_pred); plt.axhline(0, color="r", ls="--")
```

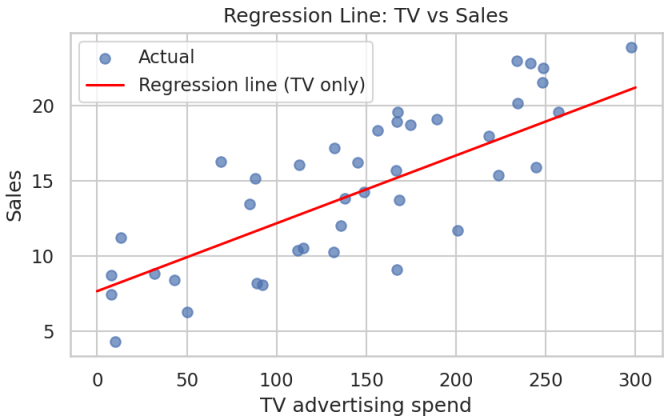


Figure 2: Regression line (TV vs Sales)

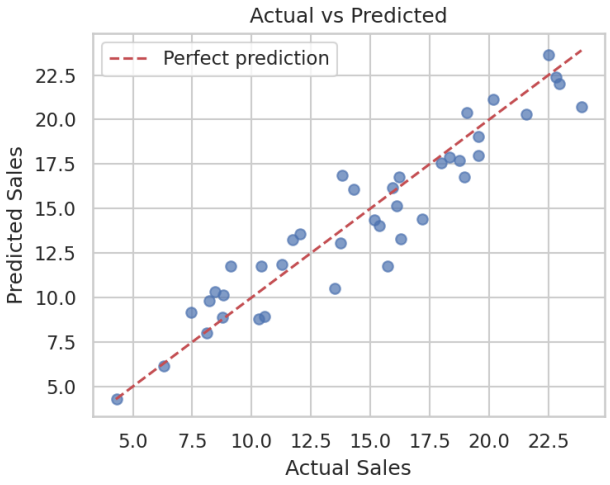


Figure 3: Actual vs predicted values (points near the dashed line = good predictions)

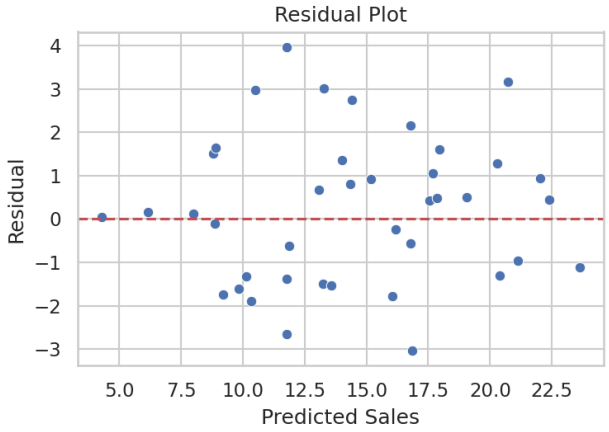


Figure 4: Residual plot (random scatter around zero = good fit)

Step 8: Evaluate the model

```
mae  = mean_absolute_error(y_test, y_pred)
mse  = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2   = r2_score(y_test, y_pred)
print(f"MAE={mae:.3f}  MSE={mse:.3f}  RMSE={rmse:.3f}  R2={r2:.3f}")
```

Metric	Value (test set)	Meaning
MAE	1.384	On average, predictions are off by about 1.38 units of sales
MSE	2.834	Average squared error (penalizes large errors)
RMSE	1.683	Typical error in the original units (sales); larger errors weigh more
R^2 Score	0.891	The model explains about 89% of the variance in Sales

Training R^2 = 0.879 and test R^2 = 0.891, which are close to each other, so there is no sign of overfitting.

Step 9: Regression coefficients

```
print("Intercept:", model.intercept_)
print(pd.DataFrame({"Feature": X.columns, "Coefficient": model.coef_}))
```

Feature	Coefficient	Interpretation
Intercept	2.720	Expected sales with zero advertising
TV	0.0468	Holding others fixed, each extra unit of TV spend raises sales by about 0.047 units
Radio	0.2029	Each extra unit of Radio spend raises sales by about 0.203 units (the biggest effect per unit)
Newspaper	-0.0062	Almost zero: Newspaper has no real impact on sales

Regression equation: Sales = 2.720 + 0.0468 x TV + 0.2029 x Radio - 0.0062 x Newspaper

Step 10: Interpretation and Suitability of Linear Regression

- R^2 of about 0.89 and a low RMSE show the model predicts sales well on unseen data.
- Radio gives the most sales per unit spent, TV also has a clear positive effect, and Newspaper is insignificant, so its budget could be moved to Radio or TV.
- The residuals are randomly scattered around zero with no strong pattern, and actual and predicted values lie close to the diagonal, so the linear assumption is reasonable here.
- Linear Regression is suitable because the target is continuous, relationships are approximately linear, and the model is simple and interpretable. Limitations: it is sensitive to outliers and multicollinearity and cannot capture non-linear effects (for example diminishing returns on advertising); in that case

polynomial regression or tree-based models can be tried.

Conclusion

A Linear Regression model was built, trained and evaluated on advertising data. It achieved good predictive performance and showed which advertising channels influence sales.

Answers to Questions

Q1. What is Predictive Analytics, and how is it used in real-world applications?

Predictive analytics uses historical data, statistical techniques and machine learning to forecast future outcomes or the likelihood of events. Applications:

- **Retail/e-commerce:** demand and sales forecasting, product recommendations.
- **Banking:** credit scoring, fraud detection.
- **Healthcare:** predicting disease risk and patient readmission.
- **Marketing:** customer churn and campaign response prediction.
- **Manufacturing:** predictive maintenance of machines.
- **Real estate:** house price estimation.

Q2. Explain the working principle of the Linear Regression algorithm.

Linear Regression models the relationship between a dependent variable y and one or more independent variables x as a straight line (or hyperplane): $\mathbf{y = b_0 + b_1x_1 + ... + b_nx_n + e}$, where b_0 is the intercept, b_i are coefficients (slopes) and e is the error. The algorithm finds the coefficients that minimize the sum of squared differences between actual and predicted values (**Ordinary Least Squares**), either by the closed-form normal equation or by gradient descent. Once trained, new values of x are put into the equation to predict y.

Q3. Differentiate between dependent and independent variables with suitable examples.

Aspect	Independent variable	Dependent variable
Role	Input / predictor / feature that is used to explain or predict	Output / response / target that is predicted
Controlled by	Chosen or observed by the analyst	Depends on the independent variables
Example 1	TV, Radio and Newspaper spending	Sales
Example 2	House area, number of bedrooms, location	House price
Example 3	Hours studied	Exam marks

Q4. Why is it necessary to split the dataset into training and testing sets?

A model must be judged on data it has not seen. If it is tested on the same data it was trained on, the score is over-optimistic because it may simply memorize the data (overfitting). The training set (e.g. 80%) is used to learn the parameters; the testing set (20%) gives an unbiased estimate of how well the model generalizes to new data. It also helps detect overfitting and compare different models fairly.

Q5. What is the significance of the R^2 Score in regression analysis?

R^2 (coefficient of determination) = 1 - (SS_{res} / SS_{tot}) is the proportion of the variance in the dependent variable explained by the model. It ranges up to 1: 1 means a perfect fit, 0 means the model is no better than predicting the mean, and it can be negative for very poor models. For example R^2 = 0.89 means 89% of the variation in sales is explained by the advertising variables. It does not show whether the model is unbiased or whether predictions are accurate in absolute terms, and it never decreases when features are added, so Adjusted R^2 is preferred with several features.

Q6. Differentiate between MAE, MSE, and RMSE. Which metric is more sensitive to large prediction errors?

Metric	Formula	Features
MAE	(1/n) Σ|y - ■|	Average absolute error; same unit as target; easy to interpret; treats all errors equally; robust to outliers
MSE	(1/n) Σ(y - ■)^2	Average squared error; unit is squared; penalizes large errors heavily; smooth and differentiable, used as a loss function
RMSE	√MSE	Same unit as target; also penalizes large errors; most commonly reported

MSE (and therefore RMSE) is more sensitive to large errors because errors are squared. MSE is the most sensitive of the three.

Q7. What assumptions should be satisfied before applying Linear Regression?

- **Linearity:** the relationship between features and target is linear.
- **Independence of errors:** observations/residuals are not correlated (no autocorrelation).
- **Homoscedasticity:** residuals have constant variance.
- **Normality of residuals:** errors are approximately normally distributed (important for inference).
- **No multicollinearity:** independent variables are not highly correlated with each other (check with VIF).
- **No significant outliers** or influential points, and a reasonably sufficient amount of data.

Q8. How can overfitting and underfitting affect the performance of a regression model?

Aspect	Overfitting	Underfitting
Meaning	Model learns noise and details of training data	Model is too simple to capture the pattern
Training error	Very low	High
Testing error	High (poor generalization)	High
Cause	Too many features, too complex model (high-degree polynomial), little data	Too few features, too simple model
Remedy	Regularization (Ridge/Lasso), feature selection, more data, cross-validation	Add relevant features, use a more complex model

Q9. Mention any three real-world applications of Linear Regression in business or industry.

- **Sales forecasting:** predicting sales from advertising spend, price and seasonality.
- **Real estate:** estimating house prices from area, location and rooms.
- **Finance/insurance:** predicting stock returns, risk or insurance premium from customer attributes.
- (Others: predicting demand for electricity, employee salary from experience.)

Q10. How can feature selection improve the accuracy and interpretability of a predictive model?

- **Accuracy:** removing irrelevant or noisy features reduces overfitting and improves generalization; removing highly correlated features reduces multicollinearity and makes coefficients stable.
- **Interpretability:** fewer variables make the model simpler to explain (for example dropping Newspaper from the model above).
- **Efficiency:** faster training and lower storage and data-collection cost.
- **Methods:** correlation analysis, p-values, Recursive Feature Elimination (RFE), Lasso regularization, and VIF.

Result: A Linear Regression model was successfully trained and evaluated, and its performance and coefficients were interpreted.